Time allowed: 3 hours 15 minutes
This question paper is divided into two sections:
Section A – This ONE question is compulsory and MUST be attempted
Section B – BOTH questions are compulsory and MUST be attempted

Question # 1

Washi Co is a large, unlisted company based in Japan and its local currency is the Japanese Yen (JPY). It manufactures industrial equipment and parts. Initially Washi Co's customers consisted of other Japanese companies, but over the last 12 years it has expanded into overseas markets and also sources its materials from around the world. The company's board of directors (BoD) believes that the strategy of overseas investments, through subsidiary companies, branches and joint ventures, has directly led to the company's substantial increase in value in the past few years.

Washi Co's BoD is considering investing in a project based in Airone, whose currency is the Airone Rand (ARD). It believes that the project will be an important addition to the company's portfolio of investments, because Washi Co does not currently have a significant presence in the part of the world where Airone is located. It is intended that the project will commence in one year's time. Details of the project are given below.

Washi Co intends to finance the project through proceeds from an agreed sale of a small European subsidiary, with any remaining funding requirement being met by additional debt finance issued in Japanese Yen. The company is due to receive the proceeds from the sale of a European subsidiary company in six months' time and it will then invest these funds in short-dated Japanese treasury bills for a further six months before they are needed for the project. Washi Co has a centralised treasury department, which hedges expected future cash flows against currency fluctuations.

Funding and financial information

The agreed proceeds from the sale of the European subsidiary company receivable in six months' time are Euro (EUR) 80 million. The BoD is concerned about a negative fluctuation in EUR/JPY rate between now and in six months when the EUR 80 million will be received. Therefore, it has asked Washi Co's treasury department to hedge the expected receipt using one of currency forwards, currency futures or exchange traded currency options. Washi Co's treasury department has obtained the following information:

	JPY per EUR 1	ARD per EUR 1
Spot	129·2–132·4	92·7–95·6
Six-month forward rate	125·3–128·6	

Currency futures (contract size EUR 125,000, quotation JPY per EUR 1)

Four-month expiry	126·9
Seven-month expiry	125·2

Currency options (contract size EUR 125,000, exercise price quotation: JPY per EUR 1, premium quotation: JPY per EUR 1)

At an exercise price of JPY 126·0 per EUR 1

	Four-month expiry	Seven-month expiry
Calls	2·3	2·6
Puts	3·4	3·8

Annualised yield on short-dated Japanese treasury bills 1·20%

Airone's annual inflation rate is 9% currently, but has fluctuated markedly in the last five years. The Japanese annual inflation rate is 1·5% and has been stable for many years.

Pato Bank has offered Washi Co the possibility of using over-the-counter options to hedge the EUR receipt instead of exchange traded currency options.

Airone project information

A member of Washi Co's finance team has produced the following estimates of the Airone project which is expected to last for four years. The estimates are based on the notes given below but not on the further information. The estimates have been checked and verified independently for their numerical accuracy.

All figures are in ARD millions

Project year	0	1	2	3	4
Sales revenue		13,000	30,800	32,300	4,500
Costs		(10,200)	(24,200)	(24,500)	(3,200)
Tax allowable depreciation		(1,000)	(1,000)	(1,000)	(1,000)
Pre-tax profits		1,800	5,600	6,800	300
Tax at 15%		(270)	(840)	(1,020)	(45)
Tax allowable depreciation		1,000	1,000	1,000	1,000
Working capital	(400)				400
Investment in buildings	(5,750)				
Investment in machinery	(4,000)				
Cash flows in ARD	(10,150)	2,530	5,760	6,780	1,655

Notes (incorporated into the estimates above):

1. The estimates are based on using the end of the first year, when the project commences, as the start of the project (year 0). The numbers are given in ARD million (m).
2. The total investment required for the project is ARD 10,150m and separated into buildings, machinery and working capital in the table above. The machinery is eligible for tax allowable depreciation on a straight-line basis and the working capital is redeemable at the end of the project.
3. The impact of inflation has been incorporated into the sales revenue and cost figures, at Airone's current annual inflation figures.
4. Corporation tax has been included based on Airone's annual rate of 15%. The tax is payable in the year that the tax liability arises.

Further information (not incorporated into the estimates above):

1. Undertaking the Airone project will result in lost sales for Washi Co. These sales would have generated a pre-tax contribution of JPY 110m in the first year of the project, rising by the Japanese rate of inflation in the following years 2 to 4 of the project.
2. The Airone project costs include components which are made in Japan by Washi Co and would be imported to the Airone project. The pre-inflation revenues generated from the sale of the components are estimated to be as follows:

   In JPY millions

Project year	1	2	3	4
Components revenue	1,200	2,400	2,500	300

   These revenues are expected to increase by the Japanese inflation rate in years 2 to 4 of the project. The contribution which Washi Co expects to earn on these components is 25% of revenue.

3. The Japanese annual corporation tax rate is 30% and tax is payable in the year that the tax liability arises. A bilateral tax treaty exists between Japan and Airone, which permits offset of overseas tax against any Japanese tax liability on overseas earnings.
4. Washi Co's finance department has estimated a cost of capital of 12% to be used as a discount rate for the project.

Required:

(a) Discuss how investing in overseas projects may enable Washi Co to gain competitive advantage over its competitors, who only invest in domestic projects. (5 marks)

(b) Discuss the advantages and drawbacks of exchange traded option contracts compared with over-the-counter options. (5 marks)

(c) Prepare a report for the board of directors of Washi Co which:

(i) Estimates the expected amount of JPY receivable under each hedge choice and the additional debt finance needed to fund the Airone project for the preferred hedge choice; (12 marks)

(ii) Estimates the net present value of the Airone project in Japanese Yen, based on the end of year one being the start of the project (year 0); (9 marks)

(iii) Evaluates the preferred hedge choice mode, the debt finance needed and whether the Airone project should be undertaken, considering both financial and non-financial factors. (9marks)

Professional marks will be awarded in part (c) for the format, structure and presentation of the report (10 marks)

Question # 2

Tisa Co is considering an opportunity to produce an innovative component which, when fitted into motor vehicle engines, will enable them to utilise fuel more efficiently. The component can be manufactured using either process Omega or process Zeta. Although this is an entirely new line of business for Tisa Co, it is of the opinion that developing either process over a period of four years and then selling the productions rights at the end of four years to another company may prove lucrative.

The annual after-tax cash flows for each process are as follows:

Process Omega

Year	**0**	**1**	**2**	**3**	**4**
After-tax cash flows ($000)	(3,800)	1,220	1,153	1,386	3,829

Process Zeta

Year	**0**	**1**	**2**	**3**	**4**
After-tax cash flows ($000)	(3,800)	643	546	1,055	5,990

Tisa Co has 10 million 50c shares trading at 180c each. Its loans have a current value of $3·6 million and an average after-tax cost of debt of 4·50%. Tisa Co's capital structure is unlikely to change significantly following the investment in either process.

Elfu Co manufactures electronic parts for cars including the production of a component similar to the one being considered by Tisa Co. Elfu Co's equity beta is 1·40, and it is estimated that the equivalent equity beta for its other activities, excluding the component production, is 1·25. Elfu Co has 400 million 25c shares in issue trading at 120c each. Its debt finance consists of variable rate loans redeemable in seven years. The loans paying interest at base rate plus 120 basis points have a current value of $96 million. It can be assumed that 80% of Elfu Co's debt finance and 75% of Elfu Co's equity finance can be attributed to other activities excluding the component production.

Both companies pay annual corporation tax at a rate of 25%. The current base rate is 3·5% and the market risk premium is estimated at 5·8%.

Required:

(a) Provide a reasoned estimate of the cost of capital that Tisa Co should use to calculate the net present value of the two processes. Include all relevant calculations. (8 marks)

(b) Calculate the internal rate of return (IRR) and the modified internal rate of return (MIRR) for Process Omega. Given that the IRR and MIRR of Process Zeta are 26·6% and 23·3% respectively, recommend which process, if any, Tisa Co should proceed with and explain your recommendation. (8 marks)

(c) Elfu Co has estimated an annual standard deviation of $800,000 on one of its other projects, based on a normal distribution of returns. The average annual return on this project is $2,200,000.

Required:

Estimate the project's Value at Risk (VAR) at a 99% confidence level for one year and over the project's life of five years. Explain what is meant by the answers obtained. (4 marks)

Professional marks (5 marks)

Question # 3

Selorne Co

Selorne Co is one of the biggest removal companies in Pauland, offering home and business removals. It has a number of long-term contracts with large businesses, although it has not won any new major contracts in the last two years. Selorne Co is listed on Pauland's stock market for smaller companies. Selorne Co is financed by a mixture of equity and short and long-term debt, but its gearing level is below the average for its sector.

Selorne Co has four executive directors, who each own 20% of the company's share capital, with the other 20% owned by external shareholders. Selorne Co has paid a constant dividend since it has been listed and its share price has risen slightly over the last three years.

Selorne Co is based in a number of the large cities and towns in Pauland and owns the majority of the sites where it is located. Many of its employees have worked for the company for a long time. Drivers of the lorries used by Selorne Co are required to have a special, heavy vehicles licence. Salary levels at Selorne Co are relatively high compared with other companies in the sector.

Chawon Co

Selorne Co is currently considering making a bid for Chawon Co, an unlisted company specialising in distribution and delivery services. Chawon Co is owned 100% by its founder, Chris Chawon. Chawon Co has built up a portfolio of small contracts over time. It has made unsuccessful bids for two larger contracts over the last 12 months, the bids being rejected primarily because Chawon Co was not felt to be big enough to be able to guarantee the level of service required.

Chawon Co is based in many of the same cities and towns where Selorne Co is located, although Chawon's premises are all rented. The drivers of Chawon's vehicles do not require a heavy vehicles licence. Chawon Co has a few long-serving employees who are mostly centre managers. Most of its drivers and staff, however, stay at Chawon Co for only a short time. Salary levels are low, although Chawon Co pays high levels of overtime and high bonuses if target profit levels are achieved. Chawon Co is highly geared, leading to recent media speculation about its financial viability.

Terms of bid for Chawon Co

In initial discussions about the acquisition, Chris Chawon indicated that he would prefer the consideration to be a share-for-share exchange, the terms being one Chawon Co share for five Selorne Co shares.

Chawon Co has 2 million $1 shares in issue, and Selorne Co has 50 million $0·50 shares in issue. Each Selorne Co share is currently trading at $6·50, which is a multiple of 8 of its free cash flow to equity. The multiple of 8 can be assumed to remain unchanged if the acquisition takes place. Chawon Co's free cash flow to equity is currently estimated at $7 million, with an expected annual growth rate of 3%, and it is expected to generate a return on equity of 15%.

Chris Chawon expects that the total free cash flows to equity of the combined company will increase by $5 million due to synergy benefits. He believes that Selorne Co will be able to win more contracts because it is larger and because it will be diversifying the services which it offers. He also believes that significant operational synergies can be achieved, pointing out the time Selorne Co drivers spend idle during the winter months when removal activity is traditionally lower. Chris Chawon believes that he can achieve the synergies if he is given management responsibility for the operational reorganisation, including dealing with the staff employment and retention issues. Chris Chawon thinks that synergies could also be achieved in central administration and in premises costs.

The chief executive and the finance director of Selorne Co are in favour of bidding for Chawon Co. However, one of the other executive directors is opposed to the bid. He is sceptical about the level of synergies which can be achieved and does not want Chris Chawon to be brought into the management of Selorne Co. He suggests that if the bid is to go ahead, it should be a cash offer rather than a share exchange. Selorne Co's chief executive has responded that Chris Chawon is likely to ask for a higher equivalent price if the purchase is for cash.

Financing the bid for Chawon Co

Selorne Co's finance director has pointed out that Selorne Co will need additional funding if Chawon Co is purchased for cash. He has suggested that there may be a number of possible sources of finance:

– A rights issue
– A fixed rate, long-term, bank loan
– A three-year, unsecured, mezzanine loan facility
– Convertible debt, with conversion rights being exercisable in five years' time

Required

(a) (i) Estimate the equity value of the combined company and the expected additional value arising from the combination of Selorne Co and Chawon Co. (6 marks)

(ii) Estimate the share of the gain from the combination created for Chris Chawon and the share of the gain created for Selorne Co's shareholders and comment on your results. (6 marks)

(b) Evaluate how reliable the estimates of the synergies for the combined company are likely to be and discuss the factors which may prevent the forecast synergies from being achieved. (5 marks)

(c) Discuss the factors which Selorne Co's board will consider when determining which source or sources of finance are chosen to finance a possible cash bid for the share capital of Chawon Co. (3 marks)

Professional marks (5 marks)